

Review

Fatty acid import into mitochondria

Janos Kerner, Charles Hoppel *

Department of Veterans Affairs Medical Center, Departments of Nutrition, Medicine and Pharmacology, Case Western Reserve University, Cleveland, OH 44106, USA

Received 6 October 1999; received in revised form 9 November 1999; accepted 9 November 1999

Abstract

The mitochondrial carnitine system plays an obligatory role in β -oxidation of long-chain fatty acids by catalyzing their transport into the mitochondrial matrix. This transport system consists of the malonyl-CoA sensitive carnitine palmitoyltransferase I (CPT-I) localized in the mitochondrial outer membrane, the carnitine:acylcarnitine translocase, an integral inner membrane protein, and carnitine palmitoyltransferase II localized on the matrix side of the inner membrane. Carnitine palmitoyltransferase I is subject to regulation at the transcriptional level and to acute control by malonyl-CoA. The N-terminal domain of CPT-I is essential for malonyl-CoA inhibition. In liver CPT-I activity is also regulated by changes in the enzyme's sensitivity to malonyl-CoA. As fluctuations in tissue malonyl-CoA content are parallel with changes in acetyl-CoA carboxylase activity, which in turn is under the control of 5'-AMP-activated protein kinase, the CPT-I/malonyl-CoA system is part of a fuel sensing gauge, turning off and on fatty acid oxidation depending on the tissue's energy demand. Additional mechanism(s) of short-term control of CPT-I activity are emerging. One proposed mechanism involves phosphorylation/dephosphorylation dependent direct interaction of cytoskeletal components with the mitochondrial outer membrane or CPT-I. We have proposed that contact sites between the outer and inner mitochondrial membranes form a microenvironment which facilitates the carnitine transport system. In addition, this system includes the long-chain acyl-CoA synthetase and porin as components. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: Fatty acid oxidation; Mitochondria; Carnitine palmitoyltransferase; Malonyl-CoA; Contact site; Porin

1. Introduction

Fatty acids represent a major energy source for the heart and skeletal muscle. In the liver, oxidation of fatty acids also serves an additional role. It produces ketone bodies which are important fuels for extrahepatic organs. The β -oxidation of activated fatty acids occurs within the mitochondrial matrix and is cata-

lyzed by the sequential action of four enzyme families (acyl-CoA dehydrogenase, enoyl-CoA hydratase, 3-hydroxyacyl-CoA dehydrogenase and 3-ketoacyl-CoA thiolase), each with different substrate specificity for short-, medium- and long-chain acyl-CoAs [1,2]. Long-chain fatty acids are activated on the mitochondrial outer membrane by the long-chain acyl-CoA synthetase (LCAS) but the mitochondrial inner membrane is not permeable to these acyl-CoAs. The carnitine dependent transport of these activated fatty acids precedes their β -oxidative chain shortening. This transport system consists of three proteins, carnitine palmitoyltransferase I (CPT-I), carnitine:acyl-

* Corresponding author. Department of Veterans Affairs Medical Center, Medical Research Service (151W), 10701 East Boulevard, Cleveland, OH 44106, USA. Fax: (216) 2298509; E-mail: clh5@po.cwru.edu

carnitine translocase (CACT) and carnitine palmitoyltransferase II (CPT-II), each with a different sub-mitochondrial localization. As a first step, acyl-CoAs formed by the catalytic action of LCAS in the mitochondrial outer membrane (MOM) are converted to acylcarnitines. This transesterification is catalyzed by CPT-I, also localized in the MOM. The reaction products, long-chain acylcarnitines, are then translocated into the mitochondrial matrix in an exchange reaction catalyzed by CACT, an integral inner membrane protein. Within the matrix the acylcarnitines are then reconverted to the respective acyl-CoAs by CPT-II, an enzyme associated with the inner leaflet of the mitochondrial inner membrane (MIM).

Although the importance of the mitochondrial carnitine system in overall fatty acid oxidation has long been recognized [3,4], significant progress in our understanding of the structure, function and regulation of the mitochondrial carnitine system has only been possible with the development and application of molecular biology tools. The scope of this presentation is to provide a brief review about our current knowledge of the individual enzymes of the carnitine dependent mitochondrial fatty acid uptake mechanism, their structural and functional organization and the role of this pathway in the regulation of overall mitochondrial fatty acid oxidation.

2. Cellular uptake and activation of long-chain fatty acids

Long-chain fatty acids represent a main energy source for many organs, especially for muscle and liver. Since most tissues contain only small amounts of storage lipids, energy production depends on a continuous supply of fatty acids, mostly from adipose tissue. Fatty acids are produced by lipolysis, transported bound to albumin in blood and taken up by tissues in a process mediated by transport proteins present in the plasma membrane [5]. Once within the cell, free fatty acids are bound to fatty acid binding proteins which are present in the cytosol in high amounts [6]. Depending on the tissue and its metabolic demand, fatty acids are either converted to triglycerides or membrane phospholipids or oxidized in the mitochondria for energy production. Before being directed into storage or membranes or oxida-

tion, fatty acids are first activated to acyl-CoAs. This reaction is catalyzed by LCAS, an enzyme abundantly present in microsomes and mitochondria [2,7]. Although LCAS serves different functions in these organelles, little is known regarding its regulation. In mitochondria, LCAS is associated with the MOM and appears to be a transmembrane protein with the active site exposed to the cytosol [8]. This orientation of LCAS gives rise to cytosolic production of long-chain acyl-CoAs, which must cross both mitochondrial boundary membranes for β -oxidation. If the orientation of CPT-I in the MOM was as proposed originally [9], e.g. the catalytic site would be exposed to the intermembrane space, the long-chain acyl-CoAs would have to cross the MOM to gain access to CPT-I. However, it is now recognized that the MOM is not freely permeable to solutes as previously thought [10]. We propose that a potential route for long-chain acyl-CoAs to cross the MOM could be the voltage dependent, anion selective channel (VDAC), also called mitochondrial porin. Porin occurs at high density in the mitochondrial outer membrane and regulates the permeability of this membrane to ions and metabolites [10,11]. Consistent with this proposed role of porin are findings in the authors' laboratory. A 22 kDa polyanion at nanomolar concentrations (polyanion is a specific inhibitor of VDAC [12]), **nearly completely inhibited palmitate and palmitoyl-CoA oxidation in rat liver and rat heart mitochondria while the oxidation of glutamate, pyruvate and palmitoylcarnitine was unaffected** [13]. If the catalytic site of CPT-I was exposed to the cytosol, as recently proposed [14,15], then long-chain acylcarnitines would be produced in the cytosol and the acylcarnitines would have to cross the MOM to gain access to the CACT. In either case, transesterification of long-chain acyl-CoAs to acylcarnitines by CPT-I commits activated fatty acids to oxidation in the mitochondrial matrix.

3. Properties and membrane localization of the enzymes of the mitochondrial carnitine system

3.1. Carnitine palmitoyltransferase I

CPT-I catalyzes the formation of long-chain acylcarnitines from activated fatty acids and free carni-

tine, thus committing them to oxidation. By virtue of its inhibition by malonyl-CoA, CPT-I represents a key regulatory site controlling the flux through β -oxidation. Consistent with its central role in mitochondrial fatty acid oxidation, the enzyme exists in at least two isoforms with significantly different kinetic and regulatory properties [16]. The liver type (L-CPT-I) displays higher affinity for carnitine and lower affinity for malonyl-CoA in contrast to the muscle isoform (M-CPT-I). The isoforms are encoded by genes localized on different chromosomes and exhibit different tissue distribution (reviewed in [16,17]). The primary structures predicted from the cDNAs show a high degree of homology with calculated molecular masses of 88 150 Da (773 amino acids) and 88 227 Da (772 amino acids) for the liver and skeletal muscle isoforms, respectively. Although neither isoform has been isolated in catalytically active form yet, the full-length cDNAs have been expressed in *Saccharomyces cerevisiae* and *Pichia pastoris*, systems that lack endogenous CPT [18–20]. **The expressed proteins displayed malonyl-CoA sensitive catalytic activity and kinetic parameters (K_m and IC_{50} for carnitine and malonyl-CoA, respectively) similar to those observed with intact mitochondria from rat liver, heart and skeletal muscle.**

The amino terminus of each CPT-I isoform includes a composite leader sequence consisting of an intermembranous space targeting signal and two 20–30-residue candidate outer membrane-spanning/stop transfer hydrophobic domains [21] which are retained in the mature protein [22]. Expression of the full-length L-CPT-I in *S. cerevisiae* resulted in outer membrane insertion with retention of both catalytic and regulatory properties (malonyl-CoA inhibition) of the enzyme [20]. Import studies of in vitro transcribed-translated proteins of different deletion and fusion constructs using isolated rat liver mitochondria and expression of chimeric CPTs in *S. cerevisiae* showed that the N-terminal domain is critical for outer membrane insertion [23].

The membrane topology of CPT-I with the malonyl-CoA binding site (cytosol) and catalytic site (intermembrane space) residing on opposite sides of the outer membrane, as originally proposed [9], has recently been challenged. In a multifaceted approach, (a) using impermeable malonyl-CoA and octanoyl-CoA (immobilized on agarose beads), (b) determin-

ing the effect of proteinase K and trypsin treatment of intact and outer membrane ruptured rat liver mitochondria on CPT-I activity and malonyl-CoA inhibition, (c) immunoblot analysis of the proteolytic fragments using anti-peptide antibodies and (d) determining the accessibility of the respective epitopes on the cytosolic and intermembrane sides of the outer membrane to anti-peptide antibodies, Fraser et al. and Zammit et al. [14,15] concluded that both sites are exposed to the cytosol. The model proposed by these authors predicts a topology of CPT-I with two transmembrane domains, separated by a stretch of 27 amino acids on the intermembrane side and both the N- (short segment) and C-termini (large segment) exposed to the cytosolic side.

Amino acid sequence alignment of L-CPT-I and M-CPT-I reveals a conserved N-terminal sequence of 124 residues with two putative transmembrane domains that is present in all known CPT-I sequences. Because of analogous hydropathy profiles and predicted secondary structure in this region, it can be assumed that M-CPT-I adopts a similar membrane topology as L-CPT-I. Recently two novel M-CPT-I mRNA species have been described which are derived from alternative splicing of M-CPT-I transcripts [21,24]. The deduced amino acid sequences of the two novel enzymes have internal deletions within their respective N-termini (putative second transmembrane domain or an adjacent region) compared with the previously identified CPT-I isoforms. Their predicted sizes are 78 kDa and 83 kDa, respectively. Since leader sequences are identical to the major M-CPT-I, the novel M-CPT-I_s should be directed to the mitochondria. However, they are likely to have a different membrane/protein topology and consequently different kinetic behavior. Expression and elucidation of the topological, catalytic and regulatory properties of these novel isoforms will provide further structural insight into the malonyl-CoA regulation of CPT-I.

The high affinity malonyl-CoA regulatory site of CPT-I has not been mapped precisely. Studies suggest that both the catalytic and regulatory sites of CPT-I are in the large carboxy-terminal domain facing the cytosol and the amino terminal domain plays a role in maintaining an optimal conformation of the enzyme required for both catalytic function and malonyl-CoA sensitivity [14,15]. Detailed studies on the

role of the N-terminal domain in malonyl-CoA inhibition have been published recently [25,26]. Expression of a series of L-CPT-I deletion mutants (nested deletions (Δ) from N-terminal amino acids 1–129; Δ 18, Δ 35, Δ 52, Δ 73, Δ 83 and Δ 129) in *P. pastoris* have shown, with the exception of Δ 129, that all expressed proteins had significant catalytic activity. Only malonyl-CoA sensitivity was affected. Although deletion of the first 18 amino acid residues was sufficient to abolish malonyl-CoA inhibition, only minimal changes in catalytic properties (e.g. saturation kinetics with respect to both substrates) were observed [25]. Further information about the amino acid residue(s) essential for malonyl-CoA sensitivity has been gained by functional analysis of expressed substitution mutants [26]. These studies demonstrated that a single amino acid substitution (glutamic acid in position 3 replaced by alanine) was as effective in the abrogation of malonyl-CoA inhibition as was the deletion of the first 18 amino acids. Substitution of histidine in position 5 with alanine had only modest effect on malonyl-CoA sensitivity.

Although these data clearly demonstrate the significance of the N-terminal domain, specifically glutamate and histidine in positions 3 and 5 in malonyl-CoA inhibition, additional residues must also contribute to malonyl-CoA sensitivity. Deletion of the first 18 amino acid residues, which are identical in both isoforms, from M-CPT-I had a less dramatic effect on malonyl-CoA sensitivity [25]. Furthermore, expression of chimeric CPT-Is in which the amino termini of L and M CPT-I isoforms were switched did not reveal the expected isoform-specific differences in malonyl-CoA sensitivity [27]. Since M- and L-CPT-I differ greatly in their sensitivity to malonyl-CoA inhibition, the expressed chimeric proteins should have reflected a ‘phenotypic’ switch if the amino-terminal domains were the sole determinants of malonyl-CoA inhibition of CPT-I.

3.2. CACT

The long-chain acylcarnitines formed by CPT-I in the mitochondrial outer membrane enter the mitochondrial matrix in exchange for free carnitine. This process is catalyzed by CACT, an integral inner membrane protein. The transporter has been purified to apparent homogeneity in catalytically active form

and its properties determined following reconstitution [28–31]. The properties of the isolated transporter are very similar to those determined on intact mitochondria [32–34]. Whether CACT is membrane-bound in mitochondria or reconstituted into liposomes it catalyzes a homologous/heterologous carnitine exchange (in addition to the slow unidirectional carnitine transport), displays increasing affinity for acylcarnitines with increasing acyl chain length and identical inhibitory properties. These kinetic data suggest the existence of a single transporter protein with a substrate specificity ranging from free carnitine to long-chain acylcarnitines. This is in contrast to the known substrate specificity of the mitochondrial carnitine acyltransferases (carnitine palmitoyl- and acetyltransferase) and implies that in addition to its role in mitochondrial fatty acid oxidation the same transporter is also engaged in other functions involving membrane transport of acyl groups (see Section 6). This may explain the severity of CACT deficiency as compared to other genetic disorders of mitochondrial fatty acid oxidation (references in [17]). Consistent with the general role of CACT in intracellular acyl traffic across membranes is its localization to mitochondria and peroxisomes [35]. Among the mitochondrial metabolite transporters the carnitine carrier is unique as the transport mechanism, in the exchange mode, displays ping-pong mechanism. As a result of this kinetic mechanism, the unidirectional transport of carnitine is slow since there is no second substrate (carnitine/acylcarnitine) to bind. This uniport mode involves a reorientation of the unloaded carrier which has been termed ‘slippage’.

The human CACT has been recently cloned and its gene mapped to chromosome 3p21.31 [36–38]. Available data suggest the CACT is the product of a single gene both in rats and humans. The full-length translation product consists of 301 amino acids with a calculated mass of 33 133 Da, which is in good agreement with the value of 32.5 kDa estimated by SDS-PAGE analysis of the purified protein. It consists of three tandem repeats, a feature common to other mitochondrial transporters. Each of the three repeated sequence motifs has been proposed to be folded into two transmembrane α -helices. The two helices of the individual elements are linked by hydrophilic loops on both sides of the lipid

bilayer with both N- and C-terminus exposed to the intermembrane space [36]. The mature transporter is N-terminally modified suggesting posttranslational modification of the carrier. The recent cloning and expression of the CACT opens the way for structure/function studies by site-directed mutagenesis [36,39]. Similarly, the partial elucidation of the structure and organization of the human CACT gene will facilitate studies relevant to the regulation of this carrier in physiological and pathophysiological conditions [40].

3.3. CPT-II

Following the translocation of long-chain acylcarnitines by CACT into the mitochondrial matrix, the carnitine esters are reconverted to their respective CoAs by CPT-II, thus completing the carnitine dependent uptake of activated fatty acids. This enzyme has been isolated in catalytically active form from a variety of tissues and its physical and kinetic properties determined [41,42]. The immunological cross-reactivity, similar apparent protein subunit (66–68 kDa) and mRNA size (approx. 2.5 kb) imply that CPT-II is the product of a single gene and is expressed as the same protein in all tissues within the body [43–45]. The full-length cDNA encodes a protein containing 658 amino acids which upon import into the mitochondria is reduced in size by 25 residues, which represent the mitochondrial targeting sequence [45]. Within the matrix, the enzyme is anchored to the inner membrane as a peripheral membrane protein. Expression of the full-length cDNA in a variety of systems produced a protein which had the same properties as the protein isolated from mitochondria [18,44,46]. CPT-II is not considered to play a regulatory role in mitochondrial fatty acid uptake. Neither in its native environment as a membrane-bound enzyme nor when isolated does CPT-II display malonyl-CoA sensitivity. Under conditions where both CPT-I mRNA and protein contents show dramatic changes, the same parameters for CPT-II are already high and remain unchanged [47–50]. The findings that neither CACT nor CPT-II seem to limit the rate of mitochondrial fatty acid uptake may be related to the preservation of mitochondrial integrity by preventing the accumulation of long-chain acyl-CoAs and acylcarnitines, which possess potent detergent-like properties.

4. Regulation of CPT-I activity

4.1. Regulation at transcriptional level

A rapid increase in CPT-I mRNA abundance and CPT-I activity was observed in liver immediately after birth, and coincided with the switch to a high-fat low-carbohydrate diet (milk). CPT-I activity remains high when rats are weaned to a high-fat diet but not when weaned to a high-carbohydrate diet [48]. Similarly, increased enzyme activities and mRNA content were found in livers of fasted and insulin dependent diabetic rats [51]. These changes in mRNA abundance coincided with changes in the glucagon/insulin ratios and were associated with elevated levels of non-esterified fatty acids and decreased glucose availability. The factors responsible for regulating CPT-I expression in liver were addressed in cultured hepatocytes by Chatelain et al. [49]. Exposure of hepatocytes to dibutyryl cAMP or long-chain fatty acids increased CPT-I mRNA levels in a dose dependent manner and were antagonized by insulin. Since the effects of dibutyryl cAMP and fatty acids were additive and inhibitors of intracellular lipolysis did not abolish the effect of dibutyryl cAMP, it was concluded that cAMP and fatty acids act through different mechanisms. 2-Bromopalmitate, which is readily activated to its non-metabolizable CoA derivative, proved to be even more potent than normal fatty acids in increasing CPT-I mRNA. Moreover, the CPT-I inhibitor tetradecylglycidic acid when added alone had no effect, but enhanced the stimulatory effect of fatty acids. It seems, therefore, that activation of fatty acids, but not their metabolism, is required for induction of CPT-I transcription. Physiological concentrations of fatty acids induced rapid CPT-I gene expression also in the pancreatic β -cell line INS-1. Fatty acid metabolism beyond long-chain acyl-CoA formation was not required for induction [52].

The transcriptional effects of fatty acids are very likely mediated through the peroxisomal proliferator-activated receptor (PPAR) [53]. Functional peroxisome proliferator response elements (PPRE) have been identified in the regulatory region of numerous genes encoding enzymes involved in lipid metabolism [54,55] and more recently also in the human muscle-type CPT-I gene [56–58]. In primary rat neonatal

cardiac myocytes, oleate induced a 4.5-fold increase in M-CPT-I mRNA levels via PPAR α [57]. Expression of the genes encoding several mitochondrial β -oxidation enzymes including very long-chain and medium-chain acyl-CoA dehydrogenase, also increased in the presence of oleate. However, mRNA encoding ATPase subunit e was not affected. Increased expression of M-CPT-I mRNA also has been shown in skeletal muscle and heart of fasted rats [58]. As with cultured neonatal cardiac myocytes, this effect was mediated via PPAR α . Increased expression levels of L-CPT-I mRNA were reported in heart of the carnitine deficient juvenile visceral steatosis mouse which normalized following carnitine administration [59]. It is possible that increased gene expression was due to elevated long-chain acyl-CoA content rather than by carnitine deficiency itself. The expression level of M-CPT-I mRNA levels was not investigated.

The expression of L-CPT-I mRNA and activity was also investigated in livers of hyperthyroid, euthyroid and hypothyroid rats. mRNA levels and activity changed in parallel, both were significantly increased in hyperthyroid and significantly decreased in hypothyroid animals [60]. The mechanisms by which triiodothyronine or cAMP stimulates transcription of CPT-I gene have not been elucidated [61].

4.2. Malonyl-CoA dependent regulation

CPT-I catalyzes the first committed step of mitochondrial fatty acid oxidation and plays a significant role in the regulation of the overall process. The mechanisms by which CPT-I exerts its control is thought to be achieved through changes in concentration of malonyl-CoA (acute or short-term control) and/or through changes in the sensitivity of CPT-I to inhibition by malonyl-CoA (long-term control).

4.2.1. Changes in tissue malonyl-CoA concentration

Since the first description of reciprocal regulation of fatty acid oxidation and synthesis by malonyl-CoA in the liver [62,63], numerous studies have shown that changes in long-chain fatty acid oxidation under physiological and pathological conditions were linked to changes in intracellular malonyl-CoA concentrations. The inverse relationship between the rate of fatty acid oxidation and tissue malonyl-CoA concentration was not restricted to lipogenic tissues,

such as the liver, but also were observed in heart and skeletal muscle which lack the capacity of de novo fatty acid synthesis. In heart for example, under conditions where fatty acid oxidation varied widely, there was no change in the IC₅₀ values for malonyl-CoA [64,65]. Rather, the actual content of malonyl-CoA decreased suggesting that changes in malonyl-CoA appeared to be the key regulatory factor in fatty acid oxidation rather than an altered sensitivity to malonyl-CoA inhibition. Similar observations have been made between rates of fatty acid oxidation and tissue malonyl-CoA content in skeletal muscle, e.g. exercise, fasting and refeeding, effects of insulin, glucagon, and denervation [66–69]. The question arises how do changes in tissue malonyl-CoA concentrations occur? If tissue malonyl-CoA concentrations play a key role in the rapid adjustment in fatty acid oxidation rates, then the reactions producing and/or disposing malonyl-CoA also must be subject to acute regulation. The formation of malonyl-CoA is catalyzed by acetyl-CoA carboxylase (ACC), which is expressed in two isoforms (ACC- α and ACC- β) encoded by two distinct genes localized on different chromosomes [70]. ACC- α predominates in lipogenic tissues while ACC- β is the predominant form in heart and skeletal muscle ([71] and references therein). The major molecular difference between the two isoforms is in the first 200 amino acid residues of ACC- β which are not present in ACC- α , and this is thought to be related to the unique function of ACC- β in heart and skeletal muscle to control mitochondrial fatty acid oxidation via CPT-I inhibition by malonyl-CoA [72]. The activity of ACC isoforms is subject to acute control by rapid covalent modification of the enzyme (inactivation and activation by phosphorylation and dephosphorylation, respectively) and by feed-forward activation by citrate and feed-back inhibition by long-chain acyl-CoA esters [73]. The phosphorylation is catalyzed by 5'-AMP-activated kinase (AMPK) which itself is subject to phosphorylation by an upstream protein kinase (AMPK-kinase) [71,74,75]. It is believed that this protein kinase cascade represents a sensitive system which acts as a cellular fuel gauge. In skeletal muscle under resting conditions the low AMP/ATP ratio would keep AMPK in its inactive (dephosphorylated) form and consequently ACC in its active dephosphorylated form resulting in relative high ma-

lonyl-CoA concentration. After onset of exercise the contractile process would lead to a rise in the ADP/ATP ratio and to an even larger rise in the AMP/ATP ratio. This would lead to activation of AMPK, phosphorylation and inactivation of ACC, and to a decrease in malonyl-CoA concentration. Similar mechanisms could operate also in cardiac muscle and may become particularly important during ischemic episodes. Evidence in favor of this hypothesis has been presented both in skeletal muscle and heart [64,76–79]. While changes in ACC activity will affect tissue malonyl-CoA levels, to substantially decrease malonyl-CoA content, a decreased rate of synthesis must be accompanied by simultaneous degradation of malonyl-CoA. Malonyl-CoA decarboxylase has been suggested as a candidate enzyme in the disposal of malonyl-CoA [80]. The presence of this enzyme has been demonstrated in several tissues of different species [80,81] and at least in rat liver it is predominantly localized within the mitochondria [82]. Data presented in [80] demonstrate malonyl-CoA decarboxylase activities in rat and rabbit heart homogenates in approx. 50-fold excess over that of ACC. Clearly, these activity ratios suggest that malonyl-CoA decarboxylase is either strictly regulated or is localized in a cell compartment not accessible to cytosolic malonyl-CoA. Supportive evidence for phosphorylation of malonyl-CoA decarboxylase has been presented for rabbit heart with the phosphorylated form being less active [80]. While the proposed role of malonyl-CoA decarboxylase in the regulation of intracellular malonyl-CoA pool is attractive, further studies are needed to substantiate this role.

4.2.2. Changes in malonyl-CoA sensitivity

Under conditions of increased fatty acid oxidation, e.g. starvation and diabetes, hepatic CPT-I becomes more active and less sensitive to inhibition by malonyl-CoA. Decreased sensitivity of the enzyme coupled with decreased tissue malonyl-CoA concentration may act synergistically to allow maximal fatty acid oxidation at any given CPT-I protein abundance [83]. The changes in malonyl-CoA sensitivity precede the increase in maximal CPT-I activity, suggesting the importance of this parameter in determining the ketogenic response of the liver during starvation. This change in malonyl-CoA sensitivity of CPT-I has only been observed with liver mitochondria but

not with heart or skeletal muscle mitochondria [84]. Although the mechanism of this desensitization of CPT-I is not known, changes in membrane lipid environment have been implicated by several laboratories ([85] and references therein). Indeed, when osmotically swollen rat liver mitochondria were presented with lipids extracted from mitochondrial outer membranes there was a significant increase of inhibition by malonyl-CoA [86]. Surprisingly, the effect of mitochondrial outer membrane lipid extract on malonyl-CoA sensitivity was mimicked by cardiolipin and phosphatidylglycerol, but not by phosphatidylcholine and phosphatidylethanolamine which are the major lipid components in the mitochondrial outer membrane. Furthermore, cardiolipin, when added to swollen intact liver mitochondria isolated from fasted rats, not only increased but completely restored the malonyl-CoA sensitivity seen in mitochondria from fed animals. Cardiolipin is only a minor component of the mitochondrial outer membrane [87,88] but it is highly enriched in mitochondrial outer and inner membrane contact sites [89], suggesting the possibility of a specific membrane microenvironment of CPT-I. Concordant with such an interpretation are recent findings by Zammit et al. [85]. Rat liver mitochondrial outer membranes isolated from fed, fasted, diabetic and starved late pregnant rats display an inverse relationship between CPT-I sensitivity and fluidity of the membrane core. However, these changes in the two parameters are not accompanied by a consistent pattern of changes in bulk or gross lipid composition.

4.3. Malonyl-CoA independent regulation

In addition to the well-established malonyl-CoA dependent control of mitochondrial fatty acid oxidation, recent data obtained on permeabilized hepatocytes and neonatal cardiac myocytes imply additional mechanism(s) by which CPT-I and thus β -oxidation is regulated.

While regulation of hepatic fatty acid oxidation through direct phosphorylation of CPT-I has been suggested as early as 1985 by Harano et al. [90] and later by Guzman and Geelen [91], this could not be confirmed in subsequent studies using a more direct approach to the question [92]. In contrast, recent studies on isolated neonatal rat cardiac

myocytes which express both CPT-I isoforms (liver and skeletal muscle) have shown that hypoxia significantly stimulated CPT-I activity which was reversed by the tyrosine phosphatase inhibitor pervanadate and by reoxygenation [93]. Kinetic analysis of CPT-I in these cardiomyocytes revealed an increased affinity for carnitine but no change in the IC_{50} value for malonyl-CoA. Treatment with DNP-etomoxiryl-CoA, a selective inhibitor of L-CPT-I, alleviated the increased affinity for carnitine suggesting that hypoxia selectively activated the liver isoform. Since the relative abundance of mRNA of the two CPT-I isoforms was not altered, the stimulation of L-CPT-I activity by hypoxia likely reflects posttranslational modification of the enzyme protein. A myocyte protein fraction immunoprecipitated using anti-phosphotyrosine antibodies was immunoblotted with antisera raised against an internal peptide of L-CPT-I and revealed the presence of an 88 kDa polypeptide consistent with the subunit size of L-CPT-I.

Another malonyl-CoA independent mechanism of acute regulation of CPT-I activity which involves direct interaction of cytoskeletal proteins with CPT-I in the mitochondrial outer membrane has been proposed recently by Velasco et al. [94]. Using permeabilized hepatocytes as an experimental model, they found that short-term incubations with effectors which influence the rate of fatty acid oxidation also coordinately affected CPT-I activity. Since cell permeabilization resulted in an almost complete loss of cellular malonyl-CoA content, the changes in CPT-I activity were very likely malonyl-CoA independent [92,95]. These observations together with the findings that the effects on CPT-I activity were only seen with permeabilized hepatocytes but not in mitochondria isolated from these cells or in disrupted cells implicated the involvement of non-diffusible components in the regulation of CPT-I [94]. Employing compounds known to interact with specific cytoskeletal proteins to modulate their polymerization state, the intermediate filament fraction (most likely cytokeratin 8 and cytokeratin 13) was identified as a likely candidate mediating this malonyl-CoA independent regulation of CPT-I activity [94,96]. In vitro reconstitution experiments using isolated cytoskeletal fractions and mitochondria demonstrated a dose dependent inhibition of mitochondrial CPT-I activity which was abrogated by the Ca^{2+} /calmodulin depen-

dent protein kinase II. Recent studies from the same laboratory provided supportive evidence for stimulation of CPT-I by 5'-AMP dependent protein kinase (AMPK) via phosphorylation of cytoskeletal components [97,98]. Based on these findings the authors proposed that intermediate filaments in their polymerized (dephosphorylated) form constrained CPT-I in a less active form, whereas depolymerization either by pharmacological agents or by phosphorylation alleviated this constraint resulting in stimulation of CPT-I activity. In this model, Ca^{2+} /calmodulin dependent protein kinase II and possibly AMPK play a central role. Phosphorylation of cytoskeletal components by Ca^{2+} /calmodulin dependent protein kinase II and/or AMPK not only would activate CPT-I but by phosphorylation (inactivation) of ACC by AMPK could reinforce CPT-I activation by decreasing tissue malonyl-CoA concentrations.

Recent studies by Sleboda et al. [99] also support a malonyl-CoA independent regulation of CPT-I involving extramitochondrial factors. These authors found that CPT-I activity decreased rapidly when hepatocytes are placed in culture (approx. 70–80% after 6 h compared to time zero). The decrease in CPT-I activity was not associated with altered malonyl-CoA sensitivity of the enzyme and was readily reversed (5–10 min) by non-metabolizable 3-thia fatty acids and to a lesser extent also by normal fatty acids when added to the culture. The stimulatory effect of 3-thia fatty acids also was observed with permeabilized hepatocytes from which most low molecular weight compounds leaked out. Since the activation of CPT-I was only marginally retained in isolated mitochondria and 3-thia fatty acids had no stimulatory effect on CPT-I in isolated mitochondria it was concluded, that the regulation of CPT-I by 3-thia fatty acids involves extramitochondrial components. The mechanism by which these non-physiological fatty acids stimulate CPT-I activity is not known.

5. The mitochondrial carnitine system: structure-function relationships

In the foregoing, we described in some detail the properties and membrane topology of the individual components of the mitochondrial carnitine system.

The reversibility of the reactions, the localization of enzymes in different membranes and the limited water solubility of intermediates would suggest an inefficient system. So how can this fatty acid transport system function as efficiently as observed in vivo? A kinetic based explanation would be that under physiological conditions the inward flow of activated fatty acids could be the result of mass action. The relative abundance of CoASH and limited availability of free carnitine in the mitochondrial matrix, coupled with the abundance of free carnitine and low content of CoASH in the cytoplasm would support this concept [100]. However, the concentration dependent flux of acyl groups by this mass action mechanism would be associated with an accumulation of intermediates in the intermembrane space. Since this pool of intermediates has not been identified, mass action alone does not account for the efficiency. Thus, diffusion of these water-insoluble intermediates does not appear sufficient to describe sharing of intermediates between consecutive steps. A physical interaction or at least close apposition of the enzymes in the outer and inner membranes would provide a structural explanation for transfer of intermediates. Recent elucidation of specific microenvironments created by fusion or apposition of the outer and inner boundary mitochondrial membranes to form contact sites has been described. Brdiczka's group has described an elegant system for energy transfer between the cytoplasm and mitochondrial matrix involving contact sites [101]. A different microenvironment for phosphatidylserine transfer and metabolism has been elucidated between the endoplasmic reticulum and contact sites ([88] and references therein). Additionally, transport of mitochondrially directed proteins from the cytoplasm involves import proteins present in the outer and inner boundary membranes that utilize contact sites for translocation into the matrix (reviewed in [102]). By analogy this structure-function role of contact sites, we proposed that contact sites between the mitochondrial outer and inner membrane could represent the microenvironment for bringing the enzymes together to facilitate fatty acid transport into the mitochondria.

To address this question initially, we previously showed that mitoplasts prepared by French press treatment of rat liver mitochondria retained dispro-

portionally more LCAS and CPT-I than the other outer membrane markers monoamine oxidase and rotenone-insensitive cytochrome *c* reductase [103]. The inner membrane integrity of these mitoplasts was not compromised as assessed by state 3 and state 4 respiratory rates with both non-lipid and lipid substrates. Moreover, the oxidation of palmitate and palmitoyl-CoA was carnitine dependent and the CPT-I activity retained by the mitoplasts was inhibited by malonyl-CoA as well as the active site directed irreversible inhibitor etomoxiryl-CoA. Our morphological observation by electron microscopy was that the outer membrane retained by the mitoplasts appeared to involve contact sites. We proposed that the outer membrane enzymes LCAS and CPT-I were associated with contact sites which accounted for the relative enrichment in mitoplasts of LCAS and CPT-I as opposed to monoamine oxidase and rotenone-insensitive cytochrome *c* reductase. Thus, contact sites would provide a structure to give a specific microenvironment to facilitate fatty acid transport, as suggested earlier [104].

Zammit and his group have also investigated the localization of the enzymes of the mitochondrial carnitine dependent fatty acid transport in contact sites [105]. They used swell/shrink treatment and sonication of rat liver mitochondria for subfractionation according to Adams et al. [106] and studied contact sites in sucrose gradient fractions identified as the intermediate-density region between the outer (rotenone-insensitive cytochrome *c* reductase) and inner (cytochrome oxidase) membranes. Analysis of CPT-I and CPT-II protein abundance by Western blotting revealed a marked enrichment of both enzymes in the contact site fraction, relative to the outer and inner membrane fraction. In a similar study, investigating the subcellular distribution of CACT they found that the transporter is enriched in the contact site fraction to a much lesser extent than CPT-I and CPT-II [35]. A possible reason for the more uniform distribution of CACT in the mitochondrial inner membrane could be that its function is not restricted to mitochondrial fatty acid oxidation. CACT is expressed as a single protein with a substrate specificity ranging from free carnitine to long-chain acylcarnitines and also participates in other functions such as buffering the mitochondrial acetyl-CoA/free CoASH ratio and shuttling chain-shortened fatty acid oxidation prod-

ucts from the peroxisomes into the mitochondria [107–109].

We have recently addressed the molecular composition of the contact site fraction obtained from Percoll-purified rat liver mitochondria using the swell/shrink technique as described [9] followed by discontinuous sucrose gradient centrifugation of the crude outer membrane fraction [110]. This procedure yielded outer membrane, contact site and inner membrane fractions which were further purified by continuous sucrose gradient. The purified fractions were analyzed for enzyme activity, protein abundance and polypeptide profile (Table 1). The polypeptide profile of the contact site fraction revealed an enrichment of two inner membrane polypeptides with apparent sizes of 53 kDa and 27 kDa, and an outer membrane polypeptide with a subunit size of 19 kDa. The nature of these polypeptides is not known and their further characterization is currently in progress. Porin is associated with all three fractions, but with a diminished apparent content in the inner membrane. LCAS and CPT-I immunoreactive proteins were only associated with outer membrane and contact sites fractions and were absent in the inner membrane fraction. In contrast to the findings of Fraser and Zammit [105], CPT-II immunoreactive protein was found only with the inner membrane but not with

contact site fractions. The distribution of enzyme activities (LCAS, CPT-I, and CPT-II) among outer membrane, contact site and inner membrane fractions followed the same pattern as the distribution of immunoreactivity.

The phospholipid cardiolipin dramatically enhanced the malonyl-CoA sensitivity of CPT-I in rat liver mitochondria [86]. Although cardiolipin was described as an inner membrane component, recently it has been demonstrated in contact sites in equal abundance compared to the inner membrane [89]. These data coupled with the structural localization of enzymes and proteins of the mitochondrial carnitine system to contact sites strongly support the notion that contact sites represent a structural organization of the mitochondrial carnitine system which would allow for an efficient uptake of activated long-chain fatty acids.

5.1. Speculation

From the above data and discussions, two proposed models for the transport of fatty acid into mitochondria are shown in Figs. 1 and 2. Using the topological data of Zammit's group detailing both the malonyl-CoA binding and catalytic site on the cytosolic surface of CPT-I [14,15], Fig. 1 depicts

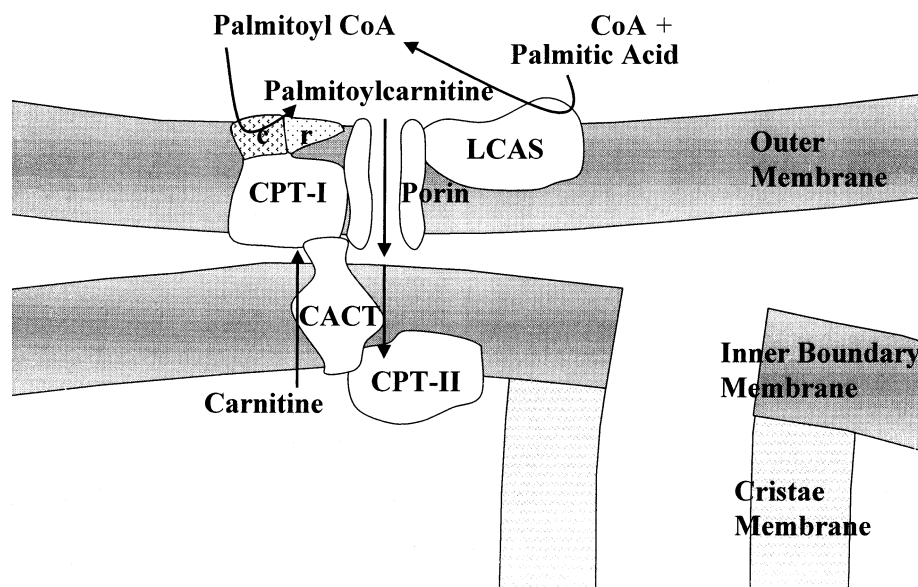


Fig. 1. Proposed model of mitochondrial fatty acid uptake with the malonyl-CoA binding site (r) and catalytic site (c) of CPT-I exposed to the cytosol as proposed by Zammit et al. [13,14]. LCAS, long chain acyl-CoA synthetase; CPT-I, malonyl-CoA sensitive carnitine palmitoyltransferase; CACT, carnitine/acylcarnitine translocase; CPT-II, malonyl-CoA insensitive carnitine palmitoyltransferase.

Table 1

Subunit sizes of polypeptides enriched in the contact site fraction and immunoblot analysis of the outer membrane, contact site and inner membrane fraction

	Outer membranes	Contact sites	Inner membranes
<i>Protein (kDa)</i>			
53	—	++	+
27	—	++	+
19	+	+	—
<i>Immunoblot analysis</i>			
MAO-B	+	+	—
LCAS	+	+	—
Porin	++	++	+
CPT-I	+	+	—
CPT-II	—	—	+

SDS-PAGE and Western blot analysis were performed as described [102]. Antisera used were commercially available (porin), provided by Dr. R. McCauley, Wayne State University, Detroit, MI (MAO-B), raised against purified enzymes (LCAS, CPT-II), or CPT-I-peptides (positions 225–241 and 620–631) coupled to keyhole limpet hemocyanin in the authors' laboratory. Distribution of enzyme activity (MAO-B, LCAS, CPT-I and CPT-II) was identical with the distribution of immunoreactive protein (data not shown). Relative abundance of the polypeptides was assigned visually.

—, absent; + < ++, present (++ differ from + by a factor of ≥ 2).

MAO-B, monoamine oxidase B; LCAS, long-chain acyl-CoA synthetase; CPT-I and CPT-II, malonyl-CoA sensitive and insensitive carnitine palmitoyltransferase, respectively.

the associated LCAS and its product with CPT-I. In this case palmitoylcarnitine is produced in the cytoplasmic compartment. Thus acylcarnitines must traverse the outer membrane to gain access to CACT. One way for this to occur would be via the outer membrane protein porin. In the site of melding of the inner and outer membrane, the negative matrix side charge has been shown to alter the selectivity of porin from anion to cation selectivity ([10] and references therein). However, our data that the high molecular weight polyanion, a putative porin inhibitor, inhibited the oxidation of palmitoyl-CoA but not palmitoylcarnitine suggested interference at a step between these two compounds [13]. These data are not consistent with the concept that CPT-I leads to cytosolic palmitoylcarnitine production since the polyanion did not inhibit palmitoylcarnitine oxidation itself. Nevertheless, polyanion has been described to transform porin from anion to cation se-

lectivity ([10] and references therein), but this still does not resolve the conflict.

Further synthesis of the information led us to propose the scheme shown in Fig. 2. In this model, LCAS produces long-chain acyl-CoA on the cytoplasmic side of the contact site. This long-chain acyl-CoA enters the outer membrane through porin in its anion selective mode. The long-chain acyl-CoA is transported to the catalytic site of CPT-I, which has its malonyl-CoA sensitivity enhanced by the juxtaposed inner membrane cardiolipin and in this microenvironment palmitoylcarnitine is synthesized with direct access to the CACT binding site with exchange for matrix carnitine across the inner membrane. The translocated long-chain acylcarnitine probably must bind to a matrix protein with presentation to CPT-II postulated to be present in the cristae membrane. An additional feature of this model is that porin, or CPT-I in the contact site, could serve as a docking site for cytoskeletal proteins. The regulatory effect of the cytoskeleton described by Velasco et al. [94] and reviewed in Section 4.3 would be transmitted via effects on contact sites.

How can porin account for these multiple functions and be an essential component of so many different types of contact site function? The presence of multiple isoforms of porin may offer an explanation for this [111,112].

6. Interorganellar acyl traffic and reverse operation of the mitochondrial carnitine system

The mitochondrial uptake of activated fatty acids requires all three components of the mitochondrial carnitine system. However, CACT, CPT-II and carnitine acetyltransferase (CAT) seem to have additional functions, namely shuttling chain-shortened fatty acids from peroxisomes into the mitochondria and buffering the mitochondrial free CoASH/acyl-CoA ratio and [109]. Additionally, the role of carnitine in conjunction with the respective carnitine acyltransferases in interorganellar acyl traffic has long been suggested [113] but convincing evidence for this function has been provided only recently. Using fibroblasts from normal and Zellweger patients (lacking peroxisomes) and mutant skin fibroblasts with established deficiency in CPT-I, CACT and CPT-II

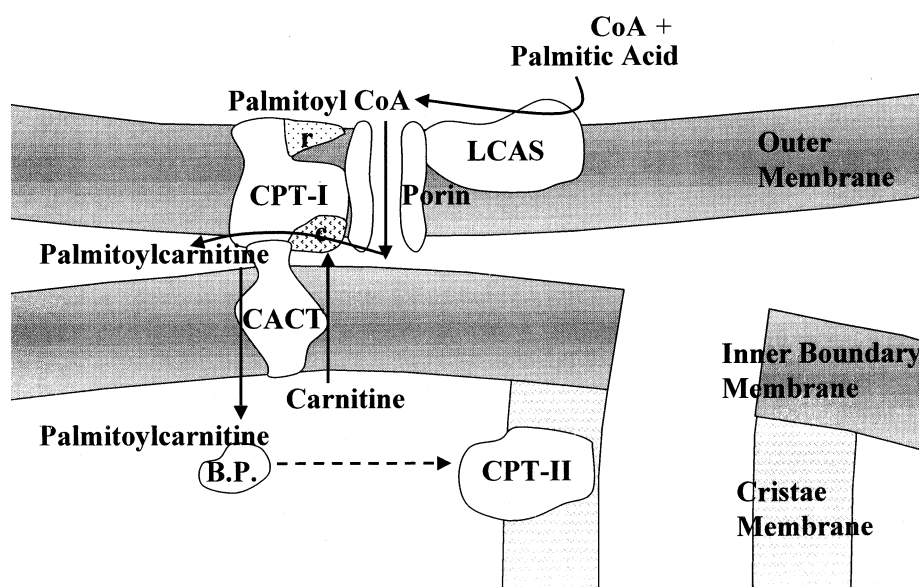


Fig. 2. Proposed model of mitochondrial fatty acid uptake with the malonyl-CoA binding site (r) of CPT-I exposed to the cytosol and the catalytic site (c) exposed to the intermembrane face of the outer membrane as proposed by Murthy and Pande [9]. B.P., postulated matrix binding protein for palmitoylcarnitine; other abbreviations are identical with those in Fig. 1.

it has been shown that the complete oxidation of phytanic and pristanic acid to CO_2 requires mitochondrial CACT and CPT-II/CAT but not CPT-I activity [108,109].

Under normal physiological conditions the direction of long-chain fatty acid movement is into the mitochondrial matrix. However, the CACT and CPT-II/CAT system can and under certain conditions does operate also in the reverse direction, e.g. removing activated short-, medium- and long-chain acyl groups out from the matrix. This has been demonstrated with isolated mitochondria, mimicking ischemic conditions [114,115] as well as with cultured fibroblasts and lymphocytes from patients with established deficiencies in mitochondrial fatty acid oxidation [116,117]. In cases of inborn errors of fatty acid oxidation, this is manifested in the appearance of specific long-chain acylcarnitines in blood and the identification of these acylcarnitines provides the diagnostic tool for the localization of the metabolic defect.

7. Future directions

In light of the pivotal role of CPT-I in overall mitochondrial fatty acid oxidation a greater under-

standing of the molecular mechanisms for malonyl-CoA and etomoxiryl-CoA inhibition, as well as isoform selectivity should provide insight for the design of drugs as inhibitors of this critical step. These drugs would be advantageous in the treatment of diabetes mellitus [118,119] and myocardial ischemia/reperfusion injury [64,79]. Furthermore, identification of the topography of the outer surface of the contact site, the organization and orientation of the putative components in the model presented in Fig. 2 should not only distinguish between the models, but establish whether contact sites provide essential structural organization for the carnitine dependent transport of acyl groups into the mitochondria. The models for the organization and pathway of acyl group transport through the contact sites reflect testable hypotheses. We believe that further elucidation of the mechanism for polyanion inhibition and the interaction of the cytoskeletal components with the contact sites will provide an in-depth look at alternative regulation of the system. It is fascinating that the peripheral benzodiazepine receptor (porin, adenine nucleotide translocase and an 18 kDa outer membrane protein) is involved in cholesterol transport to the cytochrome *P*-450 side chain cleaving enzyme localized on the outer surface of adrenal mitochondrial inner membrane [120,121]. Cholesterol

transport is the rate limiting step and the diazepam binding inhibitor greatly facilitates the side chain cleavage by increasing the transport of cholesterol through the outer membrane to the inner membrane. We are intrigued by this observation and wonder if a similar situation occurs for fatty acid transport. Could the acyl-CoA binding protein perform a similar task in enhancing acyl-CoA transport through the MOM? The homology between the diazepam binding inhibitor and acyl-CoA binding protein certainly enhances this possibility.

Acknowledgements

We thank Drs. Edward Lesnefsky and John Mieyal for reviewing the manuscript and Peter Turkaly for his help in preparing the figures and manuscript. This research was supported by the Medical Research Service of the Department of Veterans Affairs and NIH grant POI AG15885-01.

References

- [1] S. Eaton, K. Bartlett, M. Pourfarzam, Mammalian mitochondrial β -oxidation, *Biochem. J.* 320 (1996) 345–357.
- [2] W.-H. Kunau, V. Dommes, H. Schulz, β -Oxidation of fatty acids in mitochondria, peroxisomes, and bacteria: a century of continued progress, *Prog. Lipid Res.* 34 (1995) 267–342.
- [3] C.L. Hoppel, Carnitine palmitoyltransferase and transport of fatty acids, in: A. Martonosi (Ed.), *The Enzymes of Biological Membranes*, vol. 2, Plenum, New York, 1976, pp. 119–143.
- [4] J. Bremer, Carnitine – metabolism and function, *Physiol. Rev.* 63 (1983) 1420–1480.
- [5] F.A. van Nieuwenhoven, G.J. van der Vusse, J.F.C. Glatz, Membrane-associated and cytoplasmic fatty acid-binding proteins, *Lipids* 31 (1996) S223–S227.
- [6] D.A. Bernlohr, M.A. Simpson, A.V. Vogel-Hertzel, L.J. Banaszak, Intracellular lipid-binding proteins and their genes, *Annu. Rev. Nutr.* 17 (1997) 277–303.
- [7] P.A. Watkins, Fatty acid activation, *Prog. Lipid Res.* 36 (1997) 55–83.
- [8] C.B. Hesler, C. Olymbios, D. Haldar, Transverse-plane topography of long-chain acyl-CoA synthetase in the mitochondrial outer membrane, *J. Biol. Chem.* 265 (1990) 6600–6605.
- [9] M.S.R. Murthy, S.V. Pande, Malonyl-CoA binding site and the overt carnitine palmitoyltransferase activity reside on the opposite sides of the outer mitochondrial membrane, *Proc. Natl. Acad. Sci. USA* 84 (1987) 378–382.
- [10] R. Benz, Permeation of hydrophilic solutes through mitochondrial outer membranes: review on mitochondrial porins, *Biochim. Biophys. Acta* 1197 (1994) 167–196.
- [11] T. Rostovtseva, M. Colombini, ATP flux is controlled by a voltage-gated channel from the mitochondrial outer membrane, *J. Biol. Chem.* 271 (1996) 28006–28008.
- [12] T. Konig, B. Kocsis, L. Meszaros, K. Nahm, S. Zoltan, I. Horvath, Interaction of a synthetic polyanion with rat liver mitochondria, *Biochim. Biophys. Acta* 462 (1977) 380–389.
- [13] P. Turkaly, J. Kerner, C. Hoppel, A 22 kDa polyanion inhibits carnitine-dependent fatty acid oxidation in rat liver mitochondria, *FEBS Lett.* 460 (1999) 241–245.
- [14] F. Fraser, C.G. Corstorphine, V.A. Zammit, Topology of carnitine palmitoyltransferase I in the mitochondrial outer membrane, *Biochem. J.* 323 (1997) 711–718.
- [15] V.A. Zammit, F. Fraser, C.G. Corstorphine, Regulation of mitochondrial outer-membrane carnitine palmitoyltransferase (CPT I): role of membrane topology, *Adv. Enzyme Regul.* 37 (1997) 295–317.
- [16] J.D. McGarry, N.F. Brown, The mitochondrial carnitine palmitoyltransferase system, *Eur. J. Biochem.* 244 (1997) 1–14.
- [17] J. Kerner, C. Hoppel, Genetic disorders of carnitine metabolism and their nutritional management, *Annu. Rev. Nutr.* 18 (1998) 179–206.
- [18] Y. deVries, D.N. Arvidson, H.R. Waterham, J.M. Cregg, G. Woldegiorgis, Functional characterization of mitochondrial carnitine palmitoyltransferase I and II expressed in the yeast *Pichia pastoris*, *Biochemistry* 36 (1997) 5285–5292.
- [19] H. Zhu, J. Shi, Y. deVries, D.N. Arvidson, J.M. Cregg, G. Woldegiorgis, Functional studies of yeast-expressed human heart muscle carnitine palmitoyltransferase I, *Arch. Biochem. Biophys.* 347 (1997) 53–61.
- [20] C. Prip-Buus, I. Cohen, C. Kohl, V. Esser, J.D. McGarry, J. Girard, Topological and functional analysis of the rat liver carnitine palmitoyltransferase I expressed in *Saccharomyces cerevisiae*, *FEBS Lett.* 429 (1998) 173–178.
- [21] G.-S. Yu, Y.-C. Lu, T. Gulick, Expression of novel isoforms of carnitine palmitoyltransferase I (CPT-I) generated by alternative splicing of the CPT-I β gene, *Biochem. J.* 334 (1998) 225–231.
- [22] M.P. Kolodziej, V.A. Zammit, Mature carnitine palmitoyltransferase I retains the N-terminus of the nascent protein in rat liver, *FEBS Lett.* 327 (1993) 294–296.
- [23] I. Cohen, C. Kohl, J.D. McGarry, J. Girard, C. Prip-Buus, The N-terminal domain of rat liver carnitine palmitoyltransferase I mediates import into the outer mitochondrial membrane and is essential for activity and malonyl-CoA sensitivity, *J. Biol. Chem.* 273 (1998) 29896–29904.
- [24] G.-S. Yu, Y.-C. Lu, T. Gulick, Rat carnitine palmitoyltransferase I β mRNA splicing isoforms, *Biochim. Biophys. Acta* 1393 (1998) 166–172.
- [25] J. Shi, H. Zhu, D.N. Arvidson, J.M. Cregg, G. Woldegiorgis, Deletion of the conserved first 18 N-terminal amino acid residues in rat liver carnitine palmitoyltransferase I abolishes

- malonyl-CoA sensitivity and binding, *Biochemistry* 37 (1998) 11033–11038.
- [26] J. Shi, H. Zhu, D.N. Arvidson, G. Woldegiorgis, A single amino acid change (substitution of glutamate 3 with alanine) in the N-terminal region of rat liver carnitine palmitoyltransferase I abolishes malonyl-CoA inhibition and high affinity binding, *J. Biol. Chem.* 274 (1999) 9421–9426.
- [27] S.T. Swanson, D.W. Foster, J.D. McGarry, N.F. Brown, Roles of the N- and C-terminal domains of carnitine palmitoyltransferase I isoforms in malonyl-CoA sensitivity of the enzymes: insight from expression of chimaeric proteins and mutation of conserved histidine residues, *Biochem. J.* 335 (1998) 513–519.
- [28] C. Indiveri, A. Tonazzi, F. Palmieri, Identification and purification of the carnitine carrier from rat liver mitochondria, *Biochim. Biophys. Acta* 1020 (1990) 81–86.
- [29] C. Indiveri, A. Tonazzi, G. Prezioso, F. Palmieri, Kinetic characterization of the reconstituted carnitine carrier from rat liver mitochondria, *Biochim. Biophys. Acta* 1065 (1991) 231–238.
- [30] C. Indiveri, A. Tonazzi, F. Palmieri, The reconstituted carnitine carrier from rat liver mitochondria: evidence for a transport mechanism different from that of the other mitochondrial translocators, *Biochim. Biophys. Acta* 1189 (1994) 65–73.
- [31] C. Indiveri, A. Tonazzi, N. Giangregorio, F. Palmieri, Probing the active site of the reconstituted carnitine carrier from rat liver mitochondria with sulfhydryl reagents, *Eur. J. Biochem.* 228 (1995) 271–278.
- [32] S.V. Pande, R. Parvin, Characterization of carnitine acylcarnitine translocase system of heart mitochondria, *J. Biol. Chem.* 251 (1976) 6683–6691.
- [33] R. Parvin, S.V. Pande, Enhancement of mitochondrial carnitine and carnitine acylcarnitine translocase-mediated transport of fatty acids into liver mitochondria under ketogenic conditions, *J. Biol. Chem.* 254 (1979) 5423–5429.
- [34] J.A. Idell-Wenger, Carnitine:acylcarnitine translocase of rat heart mitochondria, *J. Biol. Chem.* 256 (1981) 5597–5603.
- [35] F. Fraser, V.A. Zammit, Submitochondrial and subcellular distribution of the carnitine-acylcarnitine carrier, *FEBS Lett.* 445 (1999) 41–44.
- [36] C. Indiveri, V. Iacobazzi, N. Giangregorio, F. Palmieri, The mitochondrial carnitine carrier protein: cDNA cloning, primary structure and comparison with other mitochondrial transport proteins, *Biochem. J.* 321 (1997) 713–719.
- [37] M. Huizing, V. Iacobazzi, L. Ijlst, P. Savelkoul, W. Ruitenbeek, L. van den Heuvel, C. Indiveri, J. Smeitink, F.J.M. Trijbels, R.J.A. Wanders, F. Palmieri, Cloning of the human carnitine-acylcarnitine carrier cDNA and identification of the molecular defect in a patient, *Am. J. Hum. Genet.* 61 (1997) 1239–1245.
- [38] L. Viggiano, V. Iacobazzi, R. Marzella, C. Cassano, M. Rocchi, F. Palmieri, Assignment of the carnitine/acylcarnitine translocase gene (CACT) to human chromosome band 3p21.31 by in situ hybridization, *Cytogenet. Cell Genet.* 79 (1997) 62–63.
- [39] C. Indiveri, V. Iacobazzi, N. Giangregorio, F. Palmieri, Bacterial overexpression, purification, and reconstitution of the carnitine/acylcarnitine carrier from rat liver mitochondria, *Biochem. Biophys. Res. Commun.* 249 (1998) 589–594.
- [40] V. Iacobazzi, M.A. Naglieri, C.A. Stanley, R.J.A. Wanders, F. Palmieri, The structure and organization of the human carnitine/acylcarnitine translocase (CACT) gene, *Biochem. Biophys. Res. Commun.* 252 (1998) 770–774.
- [41] L.L. Bieber, Carnitine, *Annu. Rev. Biochem.* 57 (1988) 261–283.
- [42] C.L. Hoppel, L. Brady, Carnitine palmitoyltransferase and transport of fatty acids, in: A.N. Martonosi (Ed.), *The Enzymes of Biological Membranes*, vol. 2, Plenum, New York, 1985, pp. 139–175.
- [43] K.F. Woeltje, V. Esser, B.C. Weis, W.F. Cox, J.G. Schroeder, S.-T. Liao, D.W. Foster, J.D. McGarry, Intertissue and inter-species characteristics of the mitochondrial carnitine palmitoyltransferase enzyme system, *J. Biol. Chem.* 265 (1990) 10714–10719.
- [44] K.F. Woeltje, V. Esser, B.C. Weis, A. Sen, W.F. Cox, M.J. McPhaul, C.A. Slaughter, D.W. Foster, J.D. McGarry, Cloning, sequencing, and expression of a cDNA encoding rat liver mitochondrial carnitine palmitoyltransferase II, *J. Biol. Chem.* 265 (1990) 10720–10725.
- [45] N.F. Brown, V. Esser, A.D. Gonzalez, C.T. Evans, C.A. Slaughter, D.W. Foster, J.D. McGarry, Mitochondrial import and processing of rat liver carnitine palmitoyltransferase II defines the amino terminus of the mature protein, *J. Biol. Chem.* 266 (1991) 15446–15449.
- [46] T.M. Johnson, W.R. Mann, C.J. Dragland, R.C. Anderson, G.M. Nemecek, P.A. Bell, Over-expression and characterization of active recombinant rat liver carnitine palmitoyltransferase II using baculovirus, *Biochem. J.* 309 (1995) 689–693.
- [47] G. Asins, D. Serra, G. Arias, F.G. Hegardt, Developmental changes in carnitine palmitoyltransferases I and II gene expression in intestine and liver of suckling rats, *Biochem. J.* 306 (1995) 379–384.
- [48] S. Thumelin, V. Esser, D. Charvy, M. Kolodziej, V.A. Zammit, D. McGarry, J. Girard, J.-P. Pegorier, Expression of liver carnitine palmitoyltransferase I and II genes during development in the rat, *Biochem. J.* 300 (1994) 583–587.
- [49] F. Chatelain, C. Kohl, V. Esser, J.D. McGarry, J. Girard, J.-P. Pegorier, Cyclic AMP and fatty acids increase carnitine palmitoyltransferase I gene transcription in cultured fetal rat hepatocytes, *Eur. J. Biochem.* 235 (1996) 789–798.
- [50] J.-P. Pegorier, F. Chatelain, S. Thumelin, J. Girard, Role of long-chain fatty acids in the postnatal induction of genes coding for liver mitochondrial β -oxidative enzymes, *Biochem. Soc. Trans.* 26 (1998) 113–120.
- [51] E.A. Park, R.L. Mynatt, G.D. Cook, K. Kashfi, Insulin regulates enzyme activity, malonyl-CoA sensitivity and mRNA abundance of hepatic carnitine palmitoyltransferase-I, *Biochem. J.* 310 (1995) 853–858.
- [52] F. Assimacopoulos-Jeannet, S. Thumelin, E. Roche, V. Esser, J.D. McGarry, M. Prentki, Fatty acids rapidly induce the

- carnitine palmitoyltransferase I gene in the pancreatic β -cell line INS-1, *J. Biol. Chem.* 272 (1997) 1659–1664.
- [53] T. Gulick, S. Cresci, T. Caira, D.D. Moore, D.P. Kelly, The peroxisome proliferator-activated receptor regulates mitochondrial fatty acid oxidative enzyme gene expression, *Proc. Natl. Acad. Sci. USA* 91 (1994) 11012–11016.
- [54] K. Schoonjans, B. Staels, J. Auwerx, Role of the peroxisome proliferator-activated receptor (PPAR) in mediating the effects of fibrates and fatty acids on gene expression, *J. Lipid Res.* 37 (1996) 907–925.
- [55] N. Latruffe, J. Vamecq, Peroxisome proliferators and peroxisome proliferator activated (PPARs) as regulators of lipid metabolism, *Biochimie* 79 (1997) 81–94.
- [56] C. Mascaro, E. Acosta, J.A. Ortiz, P.F. Marrero, F.G. Heghardt, D. Haro, Control of human muscle-type carnitine palmitoyltransferase I gene transcription by peroxisome proliferator-activated receptor, *J. Biol. Chem.* 273 (1998) 8560–8563.
- [57] J.M. Brandt, F. Djouadi, D.P. Kelly, Fatty acids activate transcription of the muscle carnitine palmitoyltransferase I gene in cardiac myocytes via the peroxisome proliferator-activated receptor α , *J. Biol. Chem.* 273 (1998) 23786–23792.
- [58] G.-S. Yu, Y.-C. Lu, T. Gulick, Co-regulation of tissue-specific alternative human carnitine palmitoyltransferase I β gene promoters by fatty acid enzyme substrate, *J. Biol. Chem.* 273 (1998) 32901–32909.
- [59] R. Uenaka, M. Kuwajima, A. Ono, Y. Matsuzawa, J.-I. Hayakawa, N. Inohara, Y. Kagawa, S. Ohta, Increased expression of carnitine palmitoyltransferase I gene is repressed by administering L-carnitine in the hearts of carnitine-deficient juvenile visceral steatosis mice, *J. Biochem.* 119 (1996) 533–540.
- [60] R. Mynatt, E.A. Park, F.E. Thorngate, K.H. Das, G.A. Cook, Changes in carnitine palmitoyltransferase-I mRNA abundance produced by hyperthyroidism and hypothyroidism parallel changes in activity, *Biochem. Biophys. Res. Commun.* 210 (1994) 932–937.
- [61] E.A. Park, M.L. Steffen, S. Song, V.M. Park, G.A. Cook, Cloning and characterization of the promoter for the liver isoform of the rat carnitine palmitoyltransferase I (L-CPT I) gene, *Biochem. J.* 330 (1998) 217–224.
- [62] J.D. McGarry, Y. Takabayashi, D.W. Foster, The role of malonyl-CoA in the coordination of fatty acid synthesis and oxidation in isolated rat hepatocytes, *J. Biol. Chem.* 253 (1978) 8294–8300.
- [63] J.D. McGarry, D.W. Foster, In support of the roles of malonyl-CoA and carnitine acyltransferase I in the regulation of hepatic fatty acid oxidation and ketogenesis, *J. Biol. Chem.* 254 (1979) 8163–8168.
- [64] N. Kudo, A.J. Barr, R.L. Barr, S. Desai, G.D. Lopaschuk, High rates of fatty acid oxidation during reperfusion of ischemic hearts are associated with a decrease in malonyl-CoA levels due to an increase in 5'-AMP-activated protein kinase inhibition of acetyl-CoA carboxylase, *J. Biol. Chem.* 270 (1995) 17513–17520.
- [65] G.D. Lopaschuk, L.A. Witters, T. Itoi, R. Barr, A. Barr, Acetyl-CoA carboxylase involvement in the rapid maturation of fatty acid oxidation in the newborn rabbit heart, *J. Biol. Chem.* 269 (1994) 25871–25878.
- [66] W.W. Winder, J. Arogyasami, R.J. Barton, I.M. Elayan, P.R. Hehrs, Muscle malonyl-CoA decreases during exercise, *J. Appl. Physiol.* 67 (1989) 2230–2233.
- [67] W.W. Winder, P.S. MacLean, J.C. Lucas, J.E. Fernley, G.E. Trumble, Effect of fasting and refeeding on acetyl-CoA carboxylase in rat hindlimb muscle, *J. Appl. Physiol.* 78 (1995) 578–582.
- [68] A.K. Saha, T.G. Kurowski, N.B. Ruderman, A malonyl-CoA fuel-sensing mechanism in muscle: effects of insulin, glucose, and denervation, *Am. J. Physiol.* 269 (1995) E283–E289.
- [69] N. Alam, D. Saggerson, Malonyl-CoA and the regulation of fatty acid oxidation in soleus muscle, *Biochem. J.* 334 (1998) 233–241.
- [70] K.-H. Kim, Regulation of mammalian acetyl-coenzyme A carboxylase, *Annu. Rev. Nutr.* 17 (1997) 77–99.
- [71] R.W. Brownsey, R. Zhande, A.N. Boone, Isoforms of acetyl-CoA carboxylase: structures, regulatory properties and metabolic functions, *Biochem. Soc. Trans.* 25 (1997) 1232–1238.
- [72] J. Ha, J.-K. Lee, K.-S. Kim, L.A. Witters, K.-H. Kim, Cloning of human acetyl-CoA carboxylase- β and its unique features, *Proc. Natl. Acad. Sci. USA* 93 (1996) 11466–11470.
- [73] J.B. Allred, K.E. Reilly, Short-term regulation of acetyl CoA carboxylase in tissues of higher animals, *Prog. Lipid Res.* 35 (1997) 371–385.
- [74] D.G. Hardie, D. Carling, The AMP-activated protein kinase, *Eur. J. Biochem.* 246 (1997) 259–273.
- [75] D.G. Hardie, J. Corton, Y.P. Ching, S.P. Davies, S. Hawley, Regulation of lipid metabolism by the AMP-activated protein kinase, *Biochem. Soc. Trans.* 25 (1997) 1229–1231.
- [76] W.W. Winder, D.G. Hardie, Inactivation of acetyl-CoA carboxylase and activation of AMP-activated protein kinase in muscle during exercise, *Am. J. Physiol.* 270 (1996) E299–E304.
- [77] C.A. Hutber, D.G. Hardie, W.W. Winder, Electrical stimulation inactivates muscle acetyl-CoA carboxylase and increases AMP-activated protein kinase, *Am. J. Physiol.* 272 (1997) E262–E266.
- [78] N. Kudo, J.G. Gillespie, L. Kung, L.A. Witters, R. Schulz, A.S. Clanachan, G.D. Lopaschuk, Characterization of 5'-AMP-activated protein kinase activity in the heart and its role in inhibiting acetyl-CoA carboxylase during reperfusion following ischemia, *Biochim. Biophys. Acta* 1301 (1996) 67–75.
- [79] A.-O. Makinde, J. Gamble, G.D. Lopaschuk, Upregulation of 5'-AMP-activated protein kinase is responsible for the increase in myocardial fatty acid oxidation rates following birth in the newborn rabbit, *Circ. Res.* 80 (1997) 482–489.
- [80] J.R.B. Dyck, A.J. Barr, R.L. Barr, P.E. Kolattukudy, G.D. Lopaschuk, Characterization of cardiac malonyl-CoA decarboxylase and its putative role in regulating fatty acid oxidation, *Am. J. Physiol.* 275 (1998) H2122–H2129.

- [81] H.R. Scholte, Liver malonyl-CoA decarboxylase, *Biochim. Biophys. Acta* 309 (1953) 457–465.
- [82] Y.S. Kim, P.E. Kolattukudy, Purification and properties of malonyl-CoA decarboxylase from rat liver mitochondria and its immunological comparison with the enzymes from rat brain, heart, and mammary gland, *Arch. Biochem. Biophys.* 190 (1978) 234–246.
- [83] L. Drynan, P.A. Quant, V.A. Zammit, The role of changes in the sensitivity of hepatic mitochondrial overt carnitine palmitoyltransferase in determining the onset of the ketosis of starvation in the rat, *Biochem. J.* 318 (1996) 767–770.
- [84] R.L. Mynatt, M.D. Lappi, G.A. Cook, Myocardial carnitine palmitoyltransferase of the mitochondrial outer membrane is not altered by fasting, *Biochim. Biophys. Acta* 1128 (1992) 105–111.
- [85] A.V. Zammit, C.G. Corstorphine, M.P. Kolodziej, F. Fiona, Lipid molecular order in liver mitochondrial outer membranes, and sensitivity of carnitine palmitoyltransferase I to malonyl-CoA, *Lipids* 33 (1998) 371–376.
- [86] R.L. Mynatt, J.J. Greenhaw, G.A. Cook, Cholate extracts of mitochondrial outer membranes increase inhibition by malonyl-CoA of carnitine palmitoyltransferase-I by a mechanism involving phospholipids, *Biochem. J.* 299 (1994) 761–767.
- [87] G. Daum, Lipids of mitochondria, *Biochim. Biophys. Acta* 822 (1985) 1–42.
- [88] G. Daum, J. Vance, Import of lipids into mitochondria, *Prog. Lipid Res.* 36 (1997) 103–130.
- [89] D. Ardail, J.-P. Privat, M. Egret-Charlier, C. Levrat, F. Lerme, P. Louisot, Mitochondrial contact sites, *J. Biol. Chem.* 265 (1990) 18797–18802.
- [90] Y. Harano, A. Kashiwagi, H. Kojima, M. Suzuki, T. Hashimoto, Y. Shigeta, Phosphorylation of carnitine palmitoyltransferase and activation by glucagon in isolated rat hepatocytes, *FEBS Lett.* 188 (1985) 267–272.
- [91] M. Guzman, M.J.H. Geelen, Activity of carnitine palmitoyltransferase in mitochondrial outer membranes and peroxisomes in digitonin-permeabilized hepatocytes, *Biochem. J.* 287 (1992) 487–492.
- [92] M. Guzman, M.P. Kolodziej, A. Caldwell, C.G. Corstorphine, V. Zammit, Evidence against direct involvement of phosphorylation in the activation of carnitine palmitoyltransferase by okadaic acid in rat hepatocytes, *Biochem. J.* 300 (1994) 693–699.
- [93] D. Wang, Y. Xia, L.M. Buja, J.B. McMillin, The liver isoform of carnitine palmitoyltransferase I is activated in neonatal rat cardiac myocytes by hypoxia, *Mol. Cell. Cardiol.* 180 (1998) 163–170.
- [94] G. Velasco, M.J.H. Geelen, T. Gomez del Pulgar, M. Guzman, Malonyl-CoA-independent acute control of hepatic carnitine palmitoyltransferase I activity, *J. Biol. Chem.* 273 (1998) 21497–21504.
- [95] M. Guzman, M.J.H. Geelen, Regulation of fatty acid oxidation in mammalian liver, *Biochim. Biophys. Acta* 1167 (1993) 227–241.
- [96] G. Velasco, C. Sanchez, M.J.H. Geelen, M. Guzman, Are cytoskeletal components involved in the control of hepatic carnitine palmitoyltransferase I activity?, *Biochem. Biophys. Res. Commun.* 224 (1998) 754–759.
- [97] G. Velasco, M.J.H. Geelen, M. Guzman, Control of hepatic fatty acid oxidation by 5'-AMP-activated protein kinase involves a malonyl-CoA-dependent and a malonyl-CoA-independent mechanism, *Arch. Biochem. Biophys.* 337 (1997) 169–175.
- [98] G. Velasco, T. Gomez del Pulgar, D. Carling, M. Guzman, Evidence that the AMP-activated protein kinase stimulates rat liver carnitine palmitoyltransferase I by phosphorylating cytoskeletal components, *FEBS Lett.* 439 (1998) 317–320.
- [99] J. Sleboda, K.A. Risan, O. Spydevold, J. Bremer, Short-term regulation of carnitine palmitoyltransferase I in cultured rat hepatocytes: spontaneous inactivation and reactivation by fatty acids, *Biochim. Biophys. Acta* 1436 (1999) 541–549.
- [100] J.F. Oram, J.I. Wenger, J.R. Neely, Regulation of long chain fatty acid activation in heart muscle, *J. Biol. Chem.* 250 (1975) 73–78.
- [101] D. Brdiczka, T. Walliman, The importance of the outer mitochondrial compartment in regulation of energy metabolism, *Mol. Cell. Biol.* 133/134 (1994) 69–83.
- [102] I.J. van der Klei, M. Veenhuis, W. Neupert, A morphological view on mitochondrial protein targeting, *Microsc. Res. Tech.* 27 (1994) 284–293.
- [103] C.L. Hoppel, J. Kerner, P. Turkaly, J. Turkaly, B. Tandler, The malonyl-CoA-sensitive form of carnitine palmitoyltransferase is not localized exclusively in the outer membrane of rat liver mitochondria, *J. Biol. Chem.* 273 (1998) 23495–23503.
- [104] C.L. Hoppel, J. Kerner, P. Turkaly, J. Turkaly, B. Tandler, Is carnitine palmitoyltransferase I (CPT-I) exclusively in the mitochondrial outer membrane?, *FASEB J.* 4 (1997) A445.
- [105] F. Fraser, V.A. Zammit, Enrichment of carnitine palmitoyltransferases I and II in the contact sites of rat liver mitochondria, *Biochem. J.* 329 (1998) 225–229.
- [106] V. Adams, W. Bosch, J. Schlegel, T. Walliman, D. Brdiczka, Further characterization of contact sites from mitochondria of different tissues: topology of peripheral kinases, *Biochim. Biophys. Acta* 981 (1989) 213–225.
- [107] B.S. Jacobs, R.J.A. Wanders, Fatty acid β -oxidation in peroxisomes and mitochondria: The first, unequivocal evidence for the involvement of carnitine in shuttling propionyl-CoA from peroxisomes to mitochondria, *Biochem. Biophys. Res. Commun.* 213 (1995) 1035–1041.
- [108] N.M. Verhoeven, D.S. Roe, R.M. Kok, R.J.A. Wanders, C. Jacobs, C.R. Roe, Phytanic acid and pristanic acid are oxidized by sequential peroxisomal and mitochondrial reactions in cultured fibroblasts, *J. Lipid Res.* 39 (1998) 66–74.
- [109] R.R. Ramsay, A. Arduini, The carnitine acyltransferases and their role in modulating acyl-CoA pools, *Arch. Biochem. Biophys.* 302 (1993) 307–314.
- [110] D.F. Parsons, G.R. Williams, B. Chance, Characteristics of isolated and purified preparations of the outer and inner

- membranes of mitochondria, *Ann. NY Acad. Sci.* 137 (1966) 643–666.
- [111] E. Blachly-Dyson, E.B. Zambronicz, W.H. Yu, V. Adams, E.R.B. McCabe, J. Adelman, M. Colombini, M. Forte, Cloning and functional expression in yeast of two human isoforms of the outer mitochondrial membrane channel, the voltage-dependent anion channel, *J. Biol. Chem.* 268 (1993) 1835–1841.
- [112] E. Blachly-Dyson, A. Baldini, M. Litt, E.R.B. McCabe, M. Forte, Human genes encoding the voltage-dependent anion channel (VDAC) of the outer mitochondrial membrane: mapping and identification of two new isoforms, *Genomics* 20 (1994) 62–67.
- [113] L.L. Bieber, R. Emaus, K. Valkner, S. Farrell, Possible functions of short-chain and medium-chain carnitine acyl-transferases, *Fed. Proc.* 41 (1982) 2858–2862.
- [114] N.J. Watmough, A.K.M.J. Bhuiyan, K. Bartlett, H.S.A. Sherratt, D.M. Turnbull, Skeletal muscle mitochondrial β -oxidation, *Biochem. J.* 253 (1988) 541–547.
- [115] S. Eaton, M. Purfarzam, K. Bartlett, The effect of respiratory chain impairment on β -oxidation in rat heart mitochondria, *Biochem. J.* 319 (1996) 633–640.
- [116] J. Schaefer, M. Pourfarzam, K. Bartlett, S. Jackson, D.M. Turnbull, Fatty acid oxidation in peripheral blood cells: characterization and use for the diagnosis of defects of fatty acid oxidation, *Pediatr. Res.* 37 (1995) 354–360.
- [117] M.A. Nada, W.J. Rhead, H. Sprecher, H. Schulz, C.R. Roe, Evidence for intermediate channeling in mitochondrial β -oxidation, *J. Biol. Chem.* 270 (1995) 530–535.
- [118] H.P.O. Wolf, K. Eistetter, G. Ludwig, Phenylalkyloxirane carboxylic acids, a new class of hypoglycaemic substances: hypoglycaemic and hypoketonaemic effects of sodium 2-[5-(chlorophenyl)-pentyl]-oxirane-2-carboxylate (B807-27) in fasted animals, *Diabetologia* 22 (1982) 456–463.
- [119] H.P.O. Wolf, D.W. Engel, Decrease of fatty acid oxidation, ketogenesis and gluconeogenesis in isolated perfused rat liver by phenylalkyl oxirane carboxylate (B 807-27) due to inhibition of CPT-I (EC 2.3.1.21), *Eur. J. Biochem.* 146 (1985) 359–363.
- [120] V. Papadopoulos, A.S. Brown, Role of the peripheral-type benzodiazepine receptor and the polypeptide diazepam binding inhibitor in steroidogenesis, *J. Steroid Biochem. Mol. Biol.* 53 (1995) 103–110.
- [121] M. Thomson, Molecular and cellular mechanism used in the acute phase of stimulated steroidogenesis, *Horm. Metab. Res.* 30 (1997) 16–28.